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SOCIAL SCIENCE

BAL VIGYAN

2024-25

Prepared By-

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TABLE OF CONTENTS

1. INDEX 2

2. ACKNOWLEDGEMENT 3

3. DISSERTATION APPROVAL SHEET 4

4. INTRODUCTION 5

5. IDENTIFICATION OF THE PROBLEM 6

6. AIMS AND OBJECTIVES 14

7. RESEARCH AND DATA COLLECTION 15

- Water as a Resource in India
- Land Use Patterns in India
- Tribals of India
- Project Essential - Bamboo
- Interview questionaries

8. SOCIAL INTERACTIONS 25

- Our team's visit to Tillore Khurd
- Our team's visit to Chikhli Village
- Our team's visit to Chhappan Dukaan (Nukkad Natak)

9. OUR SOLUTIONS 30
- The Bamboo Fog Catcher
- Black Soldier Flies (BSF)
- DIY Drip Kits

10. THE MODEL 37

11. TECHNOLOGY EMPHASISED 40
- The Bamboo Fog Catcher

12. CONCLUSION 49

13. BIBLIOGRAPHY 50

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- We would also like to thank all the educators in the Social Science department for their help and guidance in compiling the project. They have motivated and supported us throughout the process.

- Last but not the least, we would like to thank our parents who have directly or indirectly helped us in completing the project.

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DISSERTATION APPROVAL SHEET

This is to certify that the presented dissertation entitled “Droplets of Hope” has been completed by Ms. Jahnvi Julka, Ms. Namya Doongarwal , Ma. Arham Sethi, Ma. Yajat Khanna of “The Shishukunj International School, Jhalaria” under the guidance of the Social Science Department.

I wish them success in their work.

Principal

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INTRODUCTION

"जैसे पानी अपनी राह बनाता है, वैसे ही जनजातीय संस्कृति ने भी समय के साथ अपनी पहचान गहरी की है I"

Tribes have always been an integral to India's culture and society, representing a rich diversity of languages, traditions, and lifestyles. Comprising about 8% of the country's population, tribal communities are crucial for preserving ancient knowledge and customs.

"26.66 per cent of the total tribal population in the country did not have access to a safe source of drinking water," Ramesh Chandappa Jigajinagi, the minister of state in the Ministry of Drinking Water and Sanitation. .

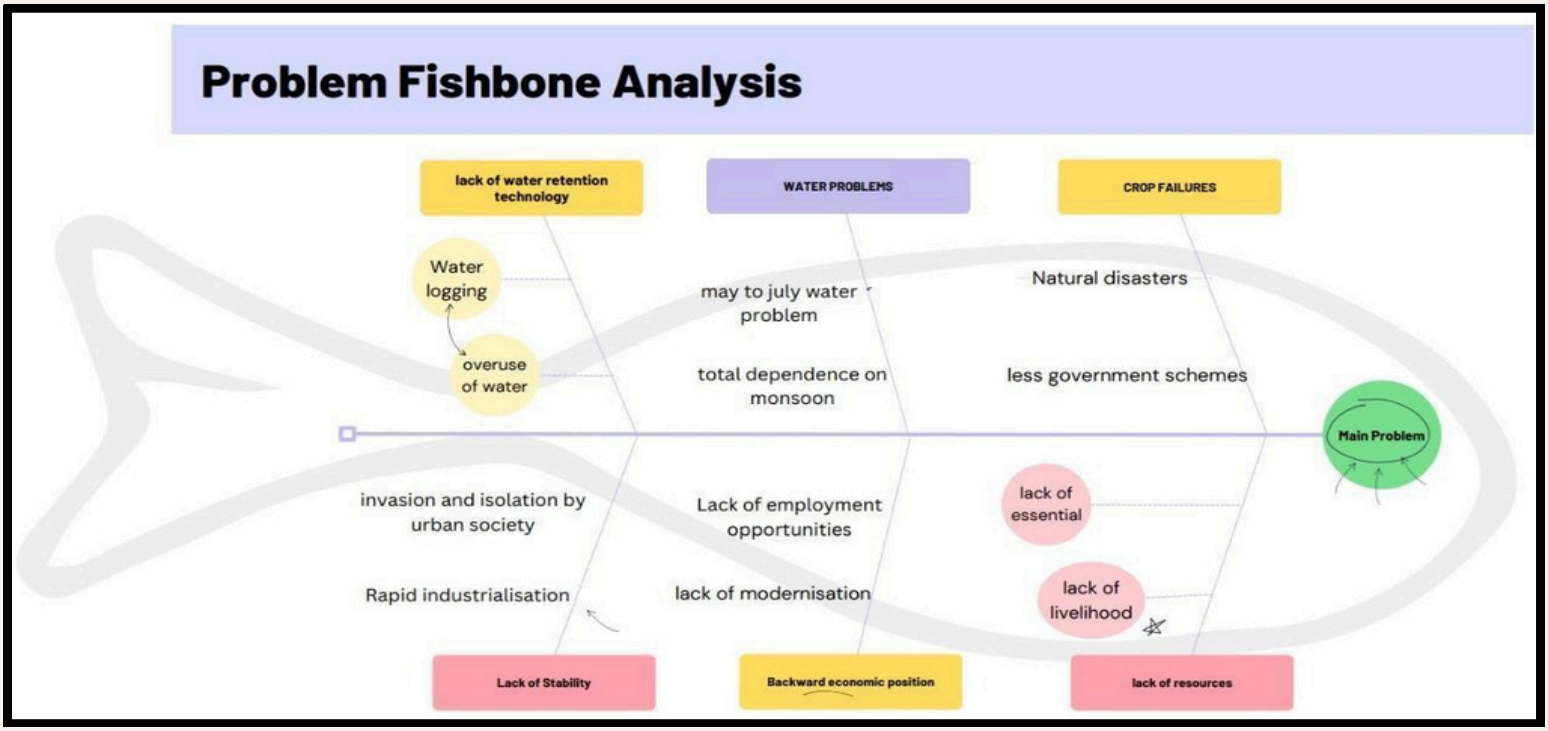
Here, sincere traditional methods of water conservation, like rainwater harvesting or spring management, are often insufficient to meet growing needs. The main water problem starts in early May and continues till July. This period of water scarcity severely impacts crop yields, as many farmers rely heavily on monsoon precipitation for irrigation.

All this isolation and lack of access to proper resources leads to lack of employment opportunities. This lack of resources, training, and access to modern technologies keeps them away from exploring new opportunities for their livelihood.

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IDENTIFICATION OF THE PROBLEM

Through our research, surveys, interviews, and questionnaires we were able to identify certain key problems that the farmers and the agricultural industries in the forested and higher elevation areas of Madhya Pradesh, we have enlisted these problems in our project as follows.



Water scarcity is a major problem for farmers in Madhya Pradesh, especially during the peak summer months of May to July. This time of minimal rainfall makes it difficult for farmers to irrigate their crops. As a result, crop yields decrease, and farmers' incomes suffer. A 2021 Central Ground Water Board report found that 84 Madhya Pradesh blocks face severe water scarcity.

Apart from the traditional concerns of water scarcity, recently the groundwater level has also rapidly declined due to irresponsible and irrational consumption of groundwater. This could lead to severe water problems in the future.

The lack of adequate water for irrigation is a major challenge for farmers in Madhya Pradesh. The government is working to address this issue by implementing various water conservation measures. However, more needs to be done to ensure that farmers have access to the water they need to grow their crops.

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WATER SCARCITY

REDUCED CROP YIELDS

When crops lack water, farmers harvest less, reducing their market sales and income.

LOWER INCOMES

Reduced sales lead to lower income, making it hard for farmers to afford necessities like food, shelter, education, and healthcare.

FOOD INSECURITY

Inadequate food production due to water scarcity can cause food insecurity, malnutrition, and health issues for farmers and their families.

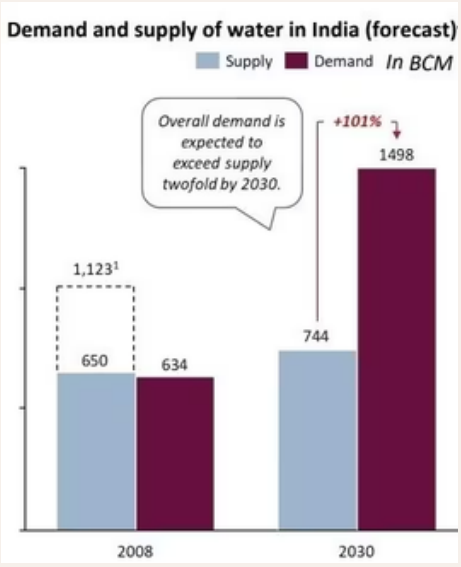
MIGRATION

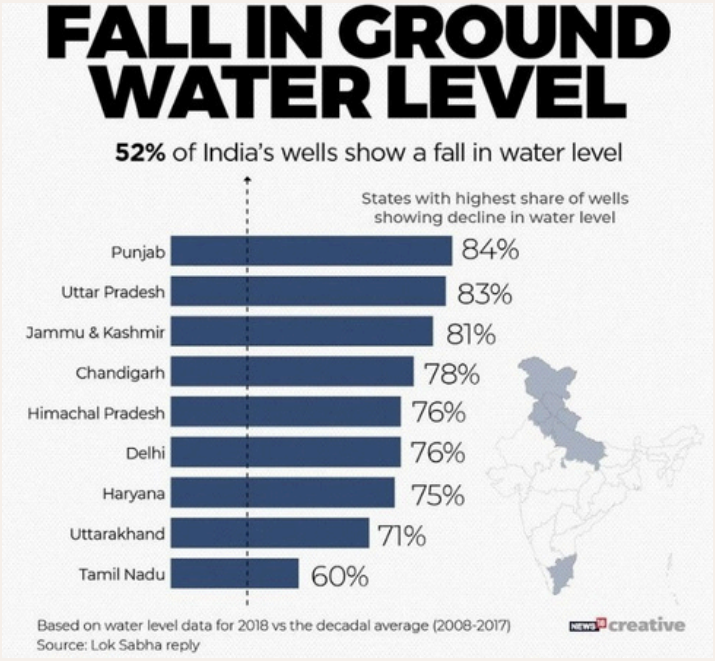
Water scarcity may force families to migrate in search of work and better opportunities, causing stress, especially for children.

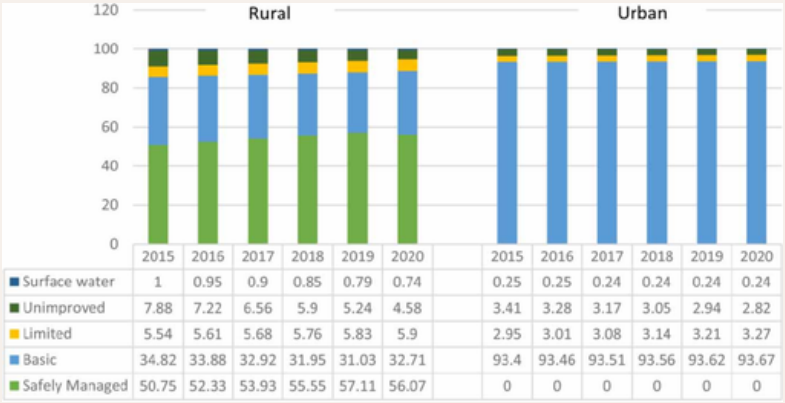
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Evidence

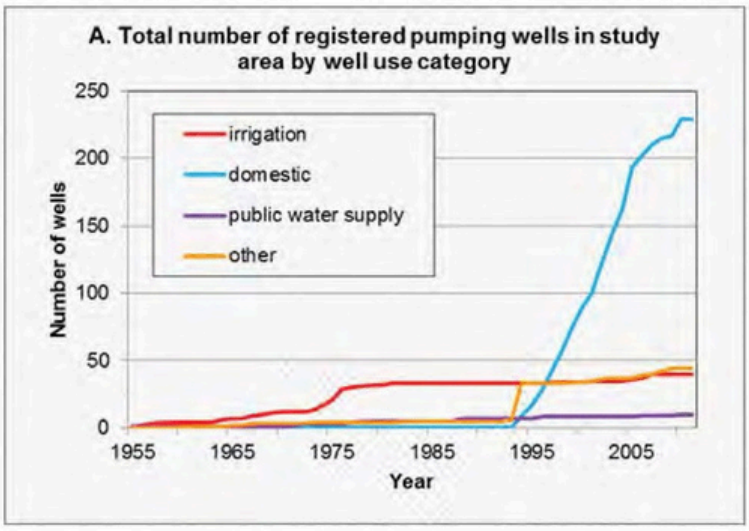
water scarcity

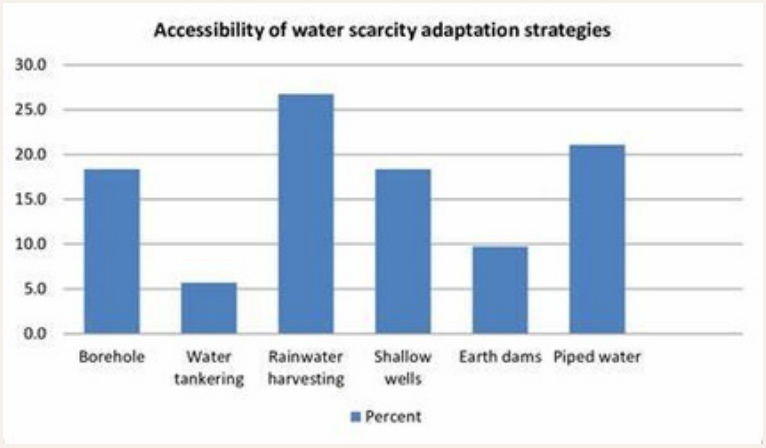


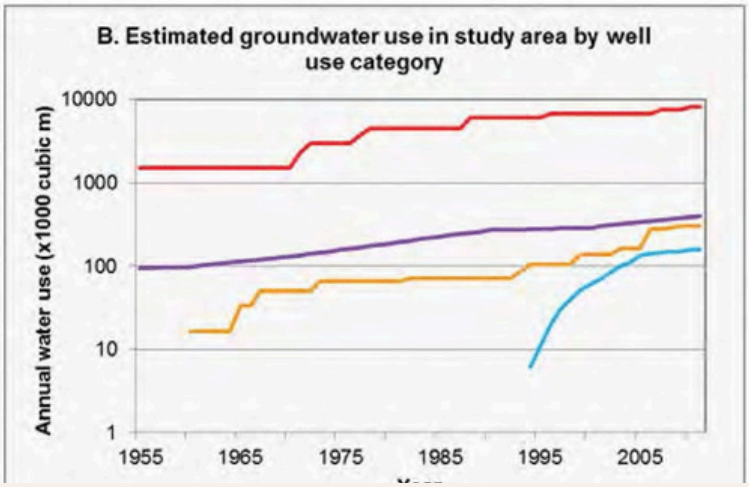


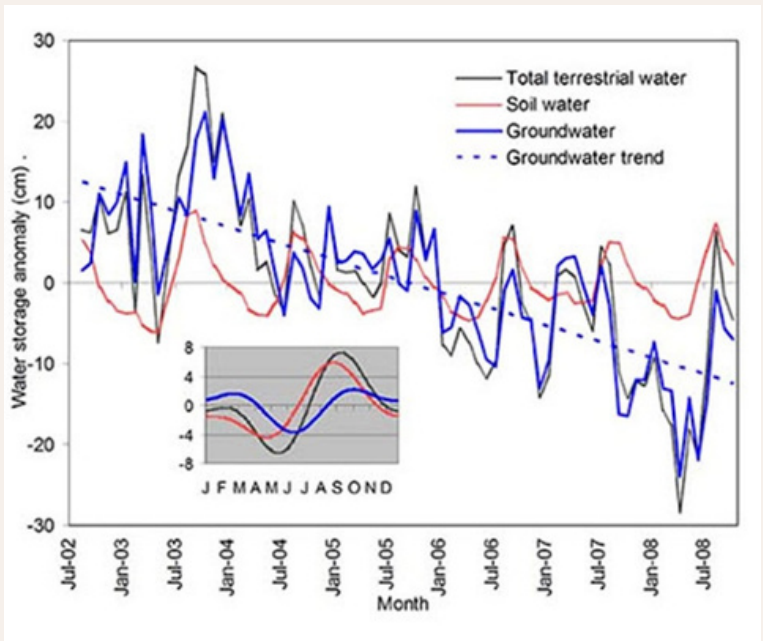


	Rural						Urban					
	2015	2016	2017	2018	2019	2020	2015	2016	2017	2018	2019	2020
Surface water	1	0.95	0.9	0.85	0.79	0.74	0.25	0.25	0.24	0.24	0.24	0.24
Unimproved	7.88	7.22	6.56	5.9	5.24	4.58	3.41	3.28	3.17	3.05	2.94	2.82
Limited	5.54	5.61	5.68	5.76	5.83	5.9	2.95	3.01	3.08	3.14	3.21	3.27
Basic	34.82	33.88	32.92	31.95	31.03	32.71	93.4	93.46	93.51	93.56	93.62	93.67
Safely Managed	50.75	52.33	53.93	55.55	57.11	56.07	0	0	0	0	0	0









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LACK OF TECHNOLOGIES

- **Water Logging**

Due to lack of technologies, water is given from underneath, which leads to oversaturation in one area leading to semi-crop failure



- **Higher cost, limited awareness**

Due to the aforementioned factors, the better technologies are available to the medium-large scale farmers (this was even reflected in our surveys)



- **Crop Failure**

The crop failure scenario is also serious due to inability to counter these factors at affordable costs



Root Rot

Excess moisture in soil promotes fungi and bacteria growth, which can damage plant roots. This prevents the plant from absorbing water and nutrients, potentially leading to its death.



- **Soil Salinity**

Soil salinity occurs when high salt content harms plant roots. Water evaporation or waterlogging can worsen salinity, making it hard for plants to absorb water and grow.



Non- Judicial use of Water

Lack of sophisticated technologies lead to overuse of water



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Evidence

lack of technology



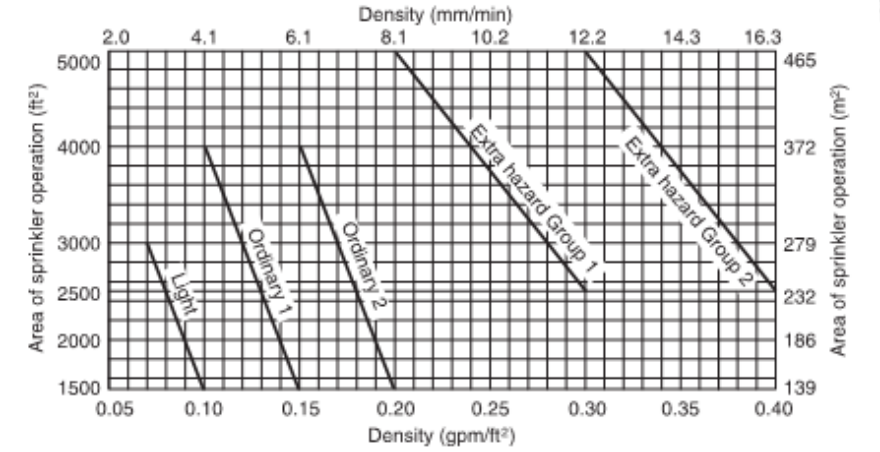
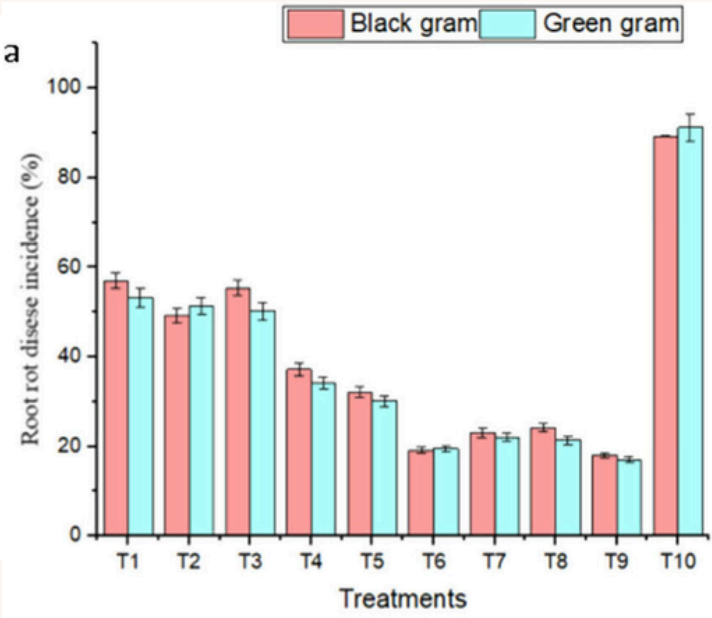
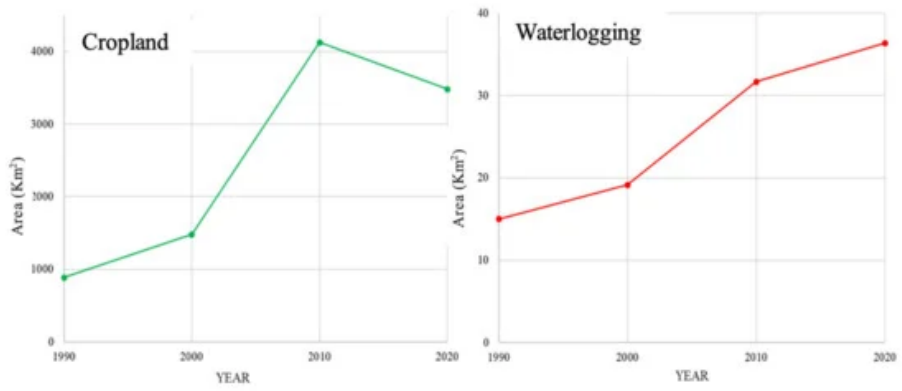
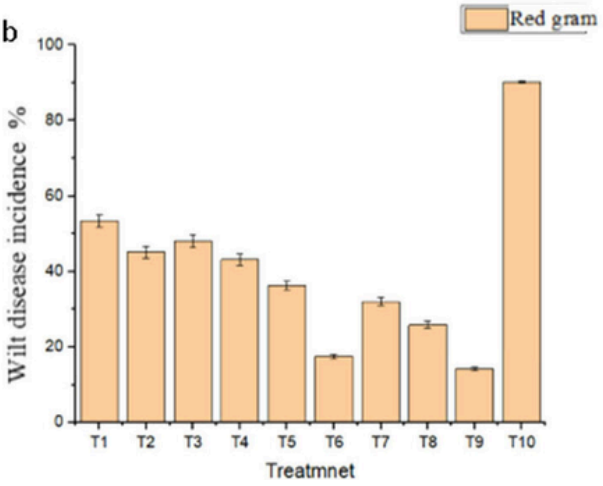
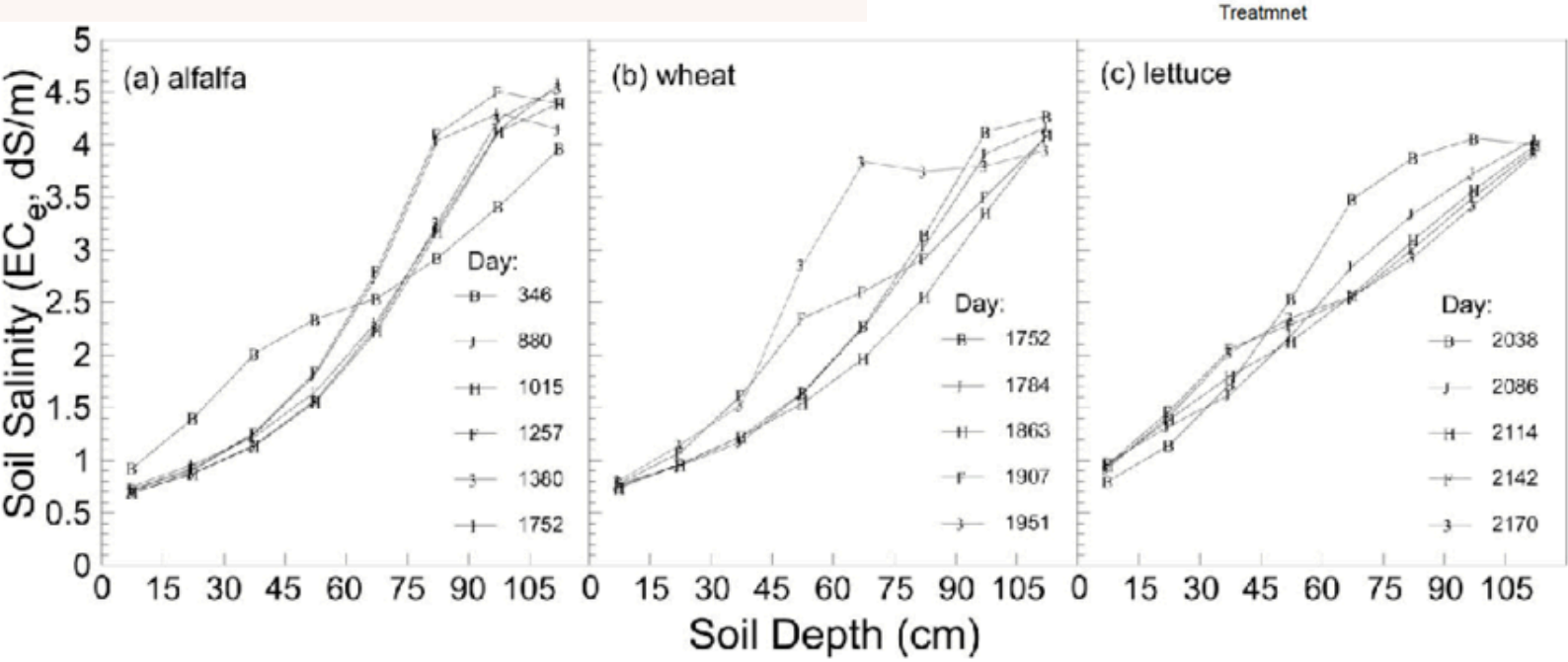


FIGURE 11.2.3.1.1 Density/Area Curves.









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SOCIO-ECONOMIC CHALLENGES IN RURAL AND TRIBAL AREAS

problem identification through social interactions

EDUCATION & SKILLS

Access to Education
- Literacy rates
- Vocational training access

Employment Opportunities
- Skill development gaps
- Low-skilled jobs

INFRASTRUCTURE GAPS

Basic Infrastructure
- Roads, electricity, healthcare

Market Access
- Limited transportation
- Credit availability

AGRICULTURAL INSTABILITY

Economic Pressures
- Input costs
- Price volatility

Environmental Factors
- Climate change (droughts, floods)
- Crop cycle instability

EMPLOYMENT & POVERTY

Employment Rates
- Unemployment rates among tribals

Economic Disadvantage
- Low-paying jobs
- Lack of socio-economic mobility





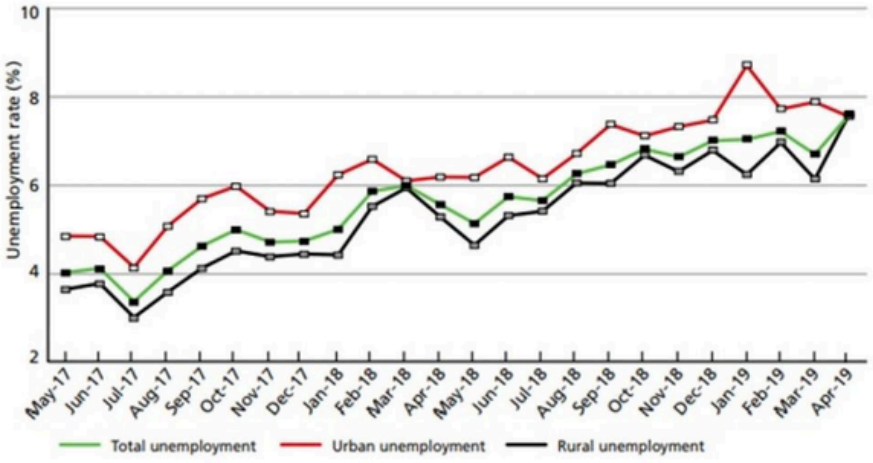
7.6%
Unemployment rate in April 2019, which was the highest in two years

1.9 times
Increase in unemployment rate in the past two years

7.61%
Rural unemployment rate in April 2019, which was the highest in two years

15
Percentage point increase in unemployment rate in youth (20-24 yrs) in past two years

Rise of the unemployed | Both rural and urban unemployment rates have risen in the past two years







EVIDENCE

socio-economic challenges in rural areas

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The state's tribal population, which accounts for around 21% of its total population, often faces limited access to education, skills training, and employment opportunities. According to the 2011 Census, the literacy rate among tribals in MP is around 50%, much lower than the state average of 70%.

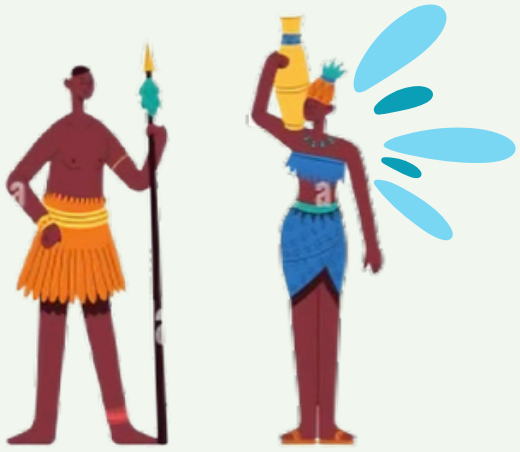
As a result, many tribal individuals remain engaged in low-paying, unskilled labour, often in agriculture or manual work. The unemployment rate in tribal regions is higher than the state average, with an estimated 7-8% of tribal youth being unemployed, compared to around 4-5% in non-tribal areas.

Many farmers in rural Madhya Pradesh rely solely on agriculture for their livelihood. However, due to factors like low agricultural productivity, erratic rainfall, and declining farm incomes, many farmers, especially small and marginal farmers, struggle to make a decent living. This lack of employment opportunities limits their income sources and makes them vulnerable to economic hardships.

The Problems

Unemployment

We landed upon the problem of unemployment which is faced by the local tribes of Madhya Pradesh in sync with conserving their culture.



Water Scarcity

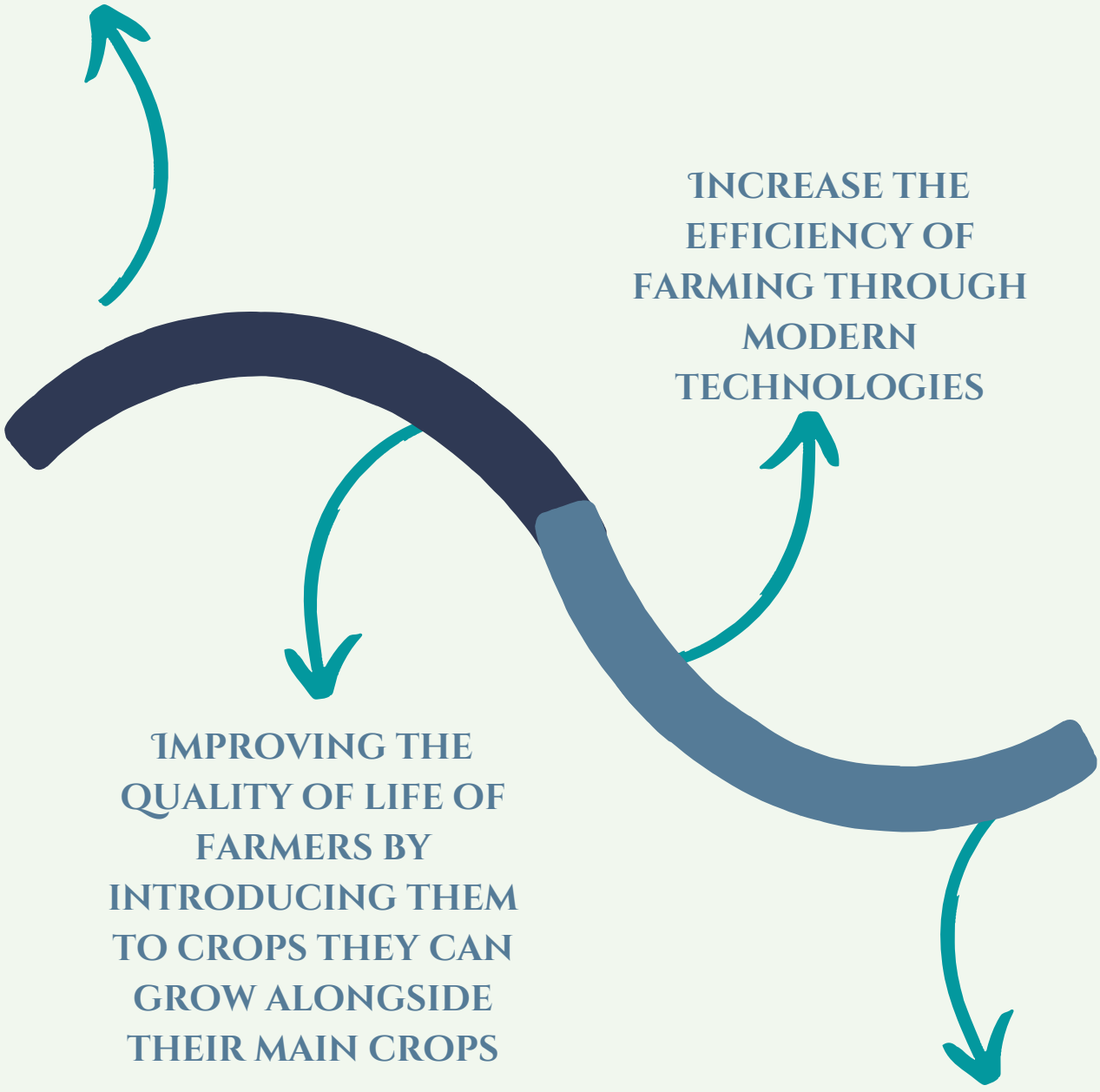
Another problem we faced was water scarcity faced by the farmers of Madhya Pradesh.

Therefore we aim to achieve this by building a connection between these two sections of society i.e. the tribal population and the farmers.

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AIMS AND OBJECTIVES

IMPROVE THE EXISTING CONDITIONS OF TRIBALS WITH RESPECT TO THE SUSTAINABLE DEVELOPMENT GOALS



INCREASE THE EFFICIENCY OF FARMING THROUGH MODERN TECHNOLOGIES

IMPROVING THE QUALITY OF LIFE OF FARMERS BY INTRODUCING THEM TO CROPS THEY CAN GROW ALONGSIDE THEIR MAIN CROPS

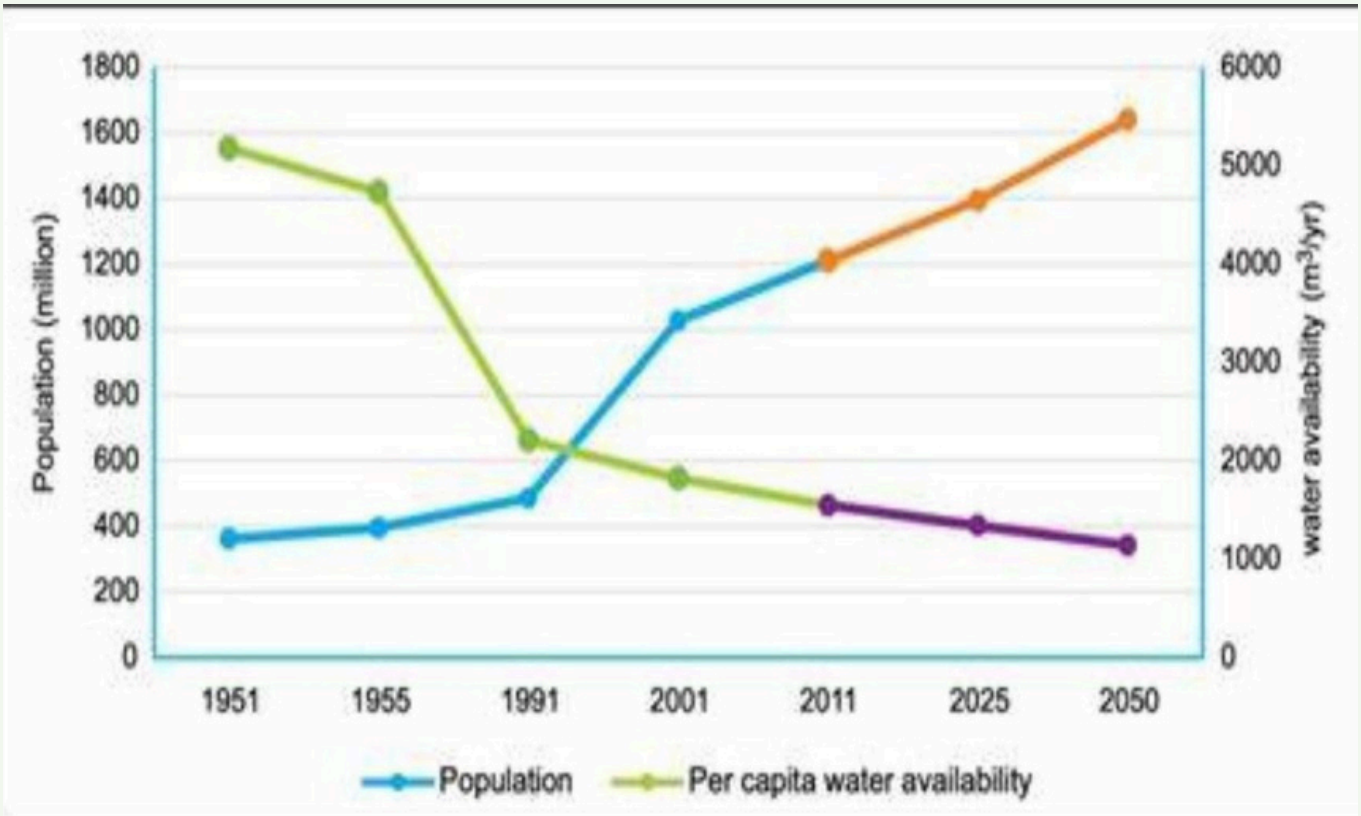
ADDRESSING THE ISSUE OF WATER SCARCITY IN OUR COUNTRY

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RESEARCH AND DATA COLLECTION

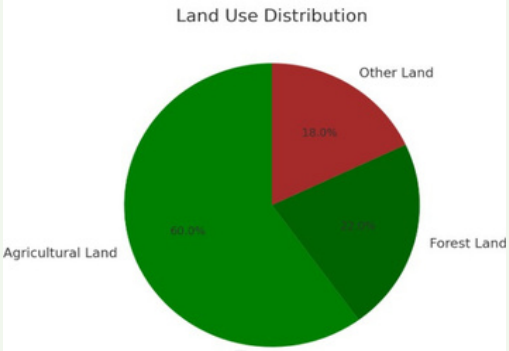
Water as a Resource in India - India has historically been an agrarian country because of its tropical and subtropical climate which allows various crops to be grown in the country. Therefore, it is given that a large share of water usage will be done to feed these farms. In 2023, India used around 784 billion cubic metres or 7.84*1014 litres of water. Out of this staggering amount, around 70% of the water was used for the agricultural sector.

On the other hand, the population of India is growing exponentially. This leads to a critical situation, where there is not enough water for everyone to use. The World Bank said, "The country has 18 per cent of the world's population, but only 4 per cent of its water resources, making it among the most water-stressed countries in the world." This situation is only predicted to worsen as the population of our country grows.



Many State governments like Tamil Nadu, have made it mandatory to install rainwater harvesting systems in all homes to ensure a water surplus. But, one serious aw in these systems is their dependence on rain, which is often unpredictable.

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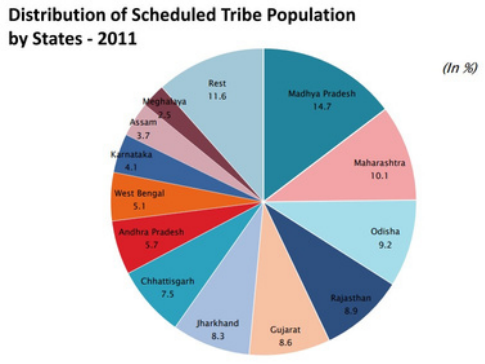


Land Use Patterns in India :

Land is a limited resource but it stands as the most important resource of any country. Land is used for all forms of economic and non-economic activities. The total land area of India is around 3.29 million km2 out of which 60% is used for agriculture and 22% is covered by forests.

This demographic representation indicates the essential role of agriculture in India. Out of the total land area of India, 13% of the land is used for bamboo farming, and the rest is used for other crops. A lot of these crops tend to get wasted due to various reasons like spoilage or low prices.

The Tribals of India: India is one of the most diverse countries in the world. It has many cultures and traditions that its people follow. Even with this statement holding true, tribal culture and people are often overlooked or prejudiced against.

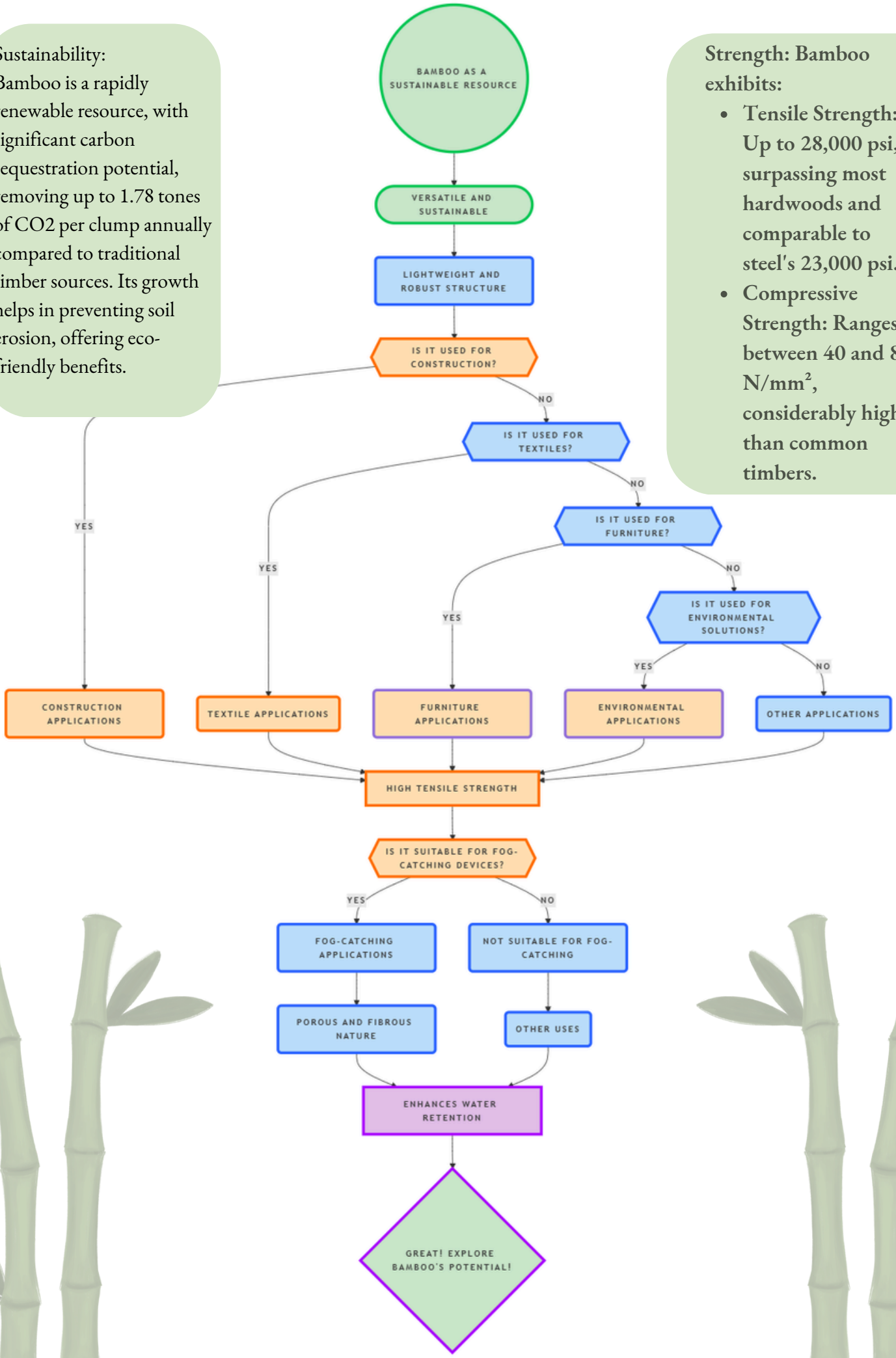


According to the 2011 Census, there are an estimated 550 unique tribes in India, which account for 8.6% of the total population of the country.

In a survey by the Planning Commission of India in 2004-05, it was noted that 47.3% of Scheduled Tribes people living in rural areas and 33.3% living in Urban areas are below the poverty line. It becomes worse in Madhya Pradesh as 58% of Rural and 45% of Urban Scheduled Tribes population are below the poverty line.

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Project Essential - Bamboo

Sustainability:
Bamboo is a rapidly renewable resource, with significant carbon sequestration potential, removing up to 1.78 tones of CO_2 per clump annually compared to traditional timber sources. Its growth helps in preventing soil erosion, offering eco-friendly benefits.



Strength: Bamboo exhibits:

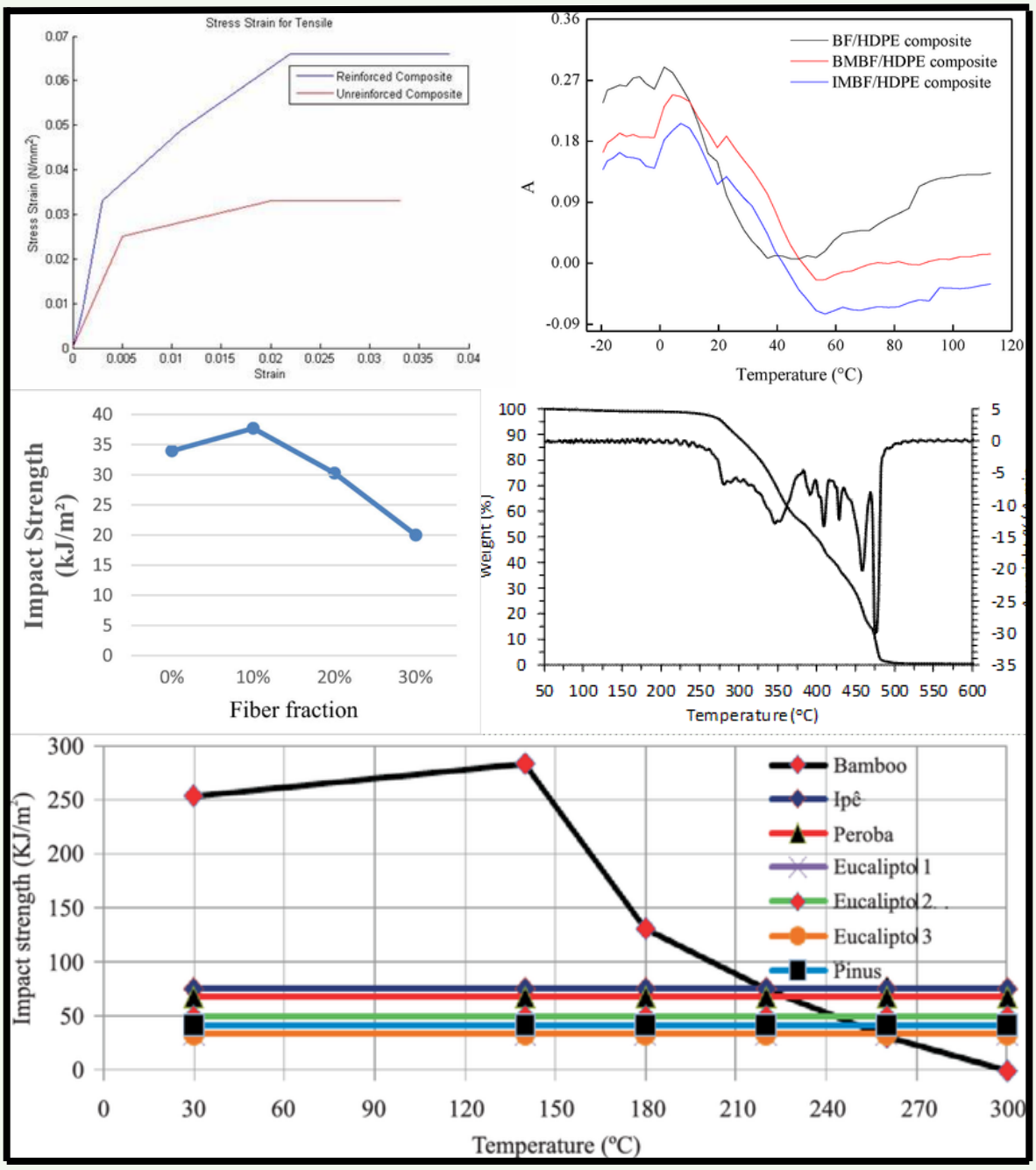
- Tensile Strength: Up to 28,000 psi, surpassing most hardwoods and comparable to steel's 23,000 psi.
- Compressive Strength: Ranges between 40 and 80 N/mm^2, considerably higher than common timbers.

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Bamboo is more environmentally friendly and cost-effective than synthetic or metal materials. Bamboo is biodegradable unlike plastics or metals, which often require energy-intensive production processes. Furthermore, its natural resilience reduces maintenance needs, making it more durable under variable weather conditions.

Its surface texture provides an ideal substrate for capturing tiny water droplets from fog. Bamboo poles can be easily combined with mesh to enhance fog collection efficiency.

Evaluation of strength, sorption, fibre etc. properties of bamboo:



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Interview Questionnaires



Questionnaire 1

(for the farmers)

Location: Tillore Khurd village

1. How much do the farmers in Tillore cultivate on Average?
तिल्लोर में किसान औसतन कितनी जमीन पर खेती करते हैं?

Ans: The area around Tillore that we cultivate ranges from 1 to 10 hectares of land.

2. Do you need any help with the water systems present, like irregular water supply or irrigation problems? How do you tackle these problems?
क्या आपको जल प्रणालियों से संबंधित किसी सहायता की आवश्यकता है, जैसे असमान जल आपूर्ति या सिंचाई की समस्याएँ? आप इन समस्याओं का समाधान कैसे करते हैं?

Ans: Yes, we face many problems with the water systems. The wells often run dry, especially during the summer months. The government pumps are unreliable and don't provide enough water for everyone. This makes it hard to irrigate our fields and grow enough crops to feed our families. Some farmers use the sprinkler system, but that is just too expensive for us.

3. Have you heard of any technologies or methods that could help you with water scarcity?
क्या आपने ऐसी किसी तकनीक या तरीके के बारे में सुना है जो पानी की कमी से निपटने में आपकी मदद कर सके?

Ans: All we have heard is about the latest drip irrigation system, which prevents excessive use of water, that method of irrigation is very effective for farmers like us. Apart from that, we have heard of the sprinkler irrigation system, which is a ordable but not that stable for small lands.

4. What solutions have you considered or implemented in your community to address water issues?
पानी की समस्याओं के समाधान के लिए आपने अपने समुदाय में किन समाधानों पर विचार किया है या उन्हें लागू किया है?

Ans: We've tried a few things to deal with the water shortage. Some farmers have dug deeper wells, but that's expensive and doesn't always work. We've also tried rainwater harvesting. This has helped during dry periods, but these too cannot sustain our fields through the farming season.

5. Are you aware of any government schemes aimed at improving water access, especially for your communities?
क्या आप ऐसी किसी सरकारी योजना के बारे में जानते हैं जिसका उद्देश्य विशेष रूप से आपके समुदायों के लिए जल पहुंच में सुधार करना है?

Ans: No, we are not aware of any such schemes implemented by the government. I don't think the government has even applied such schemes in this region. Recently the Narmada line was joined to some of the farms, but that was too only for the rich farmers. The government has supported dam formation on very small levels, which are of some help. They have also started providing information and training about Self Help Groups (SHGs) recently.

6. How can the community work with the government to better address water scarcity?
पानी की कमी को बेहतर ढंग से संबोधित करने के लिए समुदाय सरकार के साथ कैसे काम कर सकता है?

Ans: We believe that the government should create more businesses for small-scale farmers and provide regular and stable subsidies.

7. Would you be willing to grow bamboo if we helped you with it and guaranteed good pro t margins?
यदि हम इसमें आपकी मदद करें और अच्छे लाभ मार्जिन की गारंटी दें तो क्या आप बांस उगाना चाहेंगे?

Ans: Yes, we will agree to grow bamboo, provided a pre-decided term of contract.

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Questionnaire 2
(for the farmers)
Location: Chikli, Ghosikheda, and Pathan Piplya villages

1. For how many years have you been engaged in farming?
आप कितने वर्षों से खेती कर रहे हैं?

Ans: We have been practising farming since our childhood here, it is the land of our forefathers for decades.

2. Do you face any problems with the water systems present, like irregular water supply or irrigation problems?
क्या आपको मौजूदा जल प्रणालियों में किसी समस्या का सामना करना पड़ता है, जैसे अनियमित जल आपूर्ति या सिंचाई की समस्या?

Ans: Yes, we face a lot of problems with water for both drinking and irrigation. For about 8 months in a year, we hardly have any water for our fields, which often leads to a lot of crop failure. This year our soybean crops failed due to a lack of water, causing us loss. Since we have no access to the Narmada line and the tank in our village is almost empty year-round, we only have a pool to provide us with which has also been dug by us.

3. Have you heard of any technologies or methods that could help with water scarcity?
क्या आपने ऐसी किसी तकनीक या तरीके के बारे में सुना है जो पानी की कमी से निपटने में मदद कर सके?

Ans: No, currently we haven't heard of any method that could help us with our problem. We have to make do with the little water we get.

4. Are you aware of any government schemes aimed at improving water access in your community?
क्या आप अपने समुदाय में पानी की पहुंच में सुधार लाने के उद्देश्य से किसी सरकारी योजना के बारे में जानते हैं?

Ans: No, we aren't aware of these schemes at all, and we only know very little about them, just what is available to us on our phones these days. Whenever possible, we try to get the maximum subsidy we can from these schemes, we even fill out the application form. However, we get rejected several times or if our application is accepted no further information is provided to us, and we are thus not able to utilise them.



8. Do you think fog catchers could be implemented in your area?
क्या आपको लगता है कि फॉग कैचर को आपके यहां लागू किया जा सकता है?

Ans: Yes, we think this method could be extremely beneficial for us because in our village, every morning, especially in these winter months, a lot of fog, or "kohra," is present. If this fog could be condensed to get water, it would be very helpful for our farms, especially in summer.

13. Will you be willing to grow bamboo, provided we help you with it and guarantee good profit margins?
क्या आप बांस उगाने के इच्छुक होंगे, बशर्ते हम इसमें आपकी मदद करें और अच्छे लाभ मार्जिन की गारंटी दें?
Ans: Yes, we all are already growing bamboo in some parts of the village, so it would be easy for us to grow more and expand this stream of farming as well. Near the village further, there is a "Ghati" where multiple bamboos grow.

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Questionnaire 3

(Dr. Kamlesh Kumar, telephonic)

Location: Samdarshi Seva Kendra NGO

1. Which regions do you operate in, and which communities do you mainly support?

Ans: We primarily focus on working with Scheduled Tribes (ST) and Scheduled Castes (SC), supporting several tribes and castes across these communities. Our primary region of operation is Madhya Pradesh, but we also extend our efforts into Chhattisgarh. Currently, however, many of our projects, including those funded by the Central Government, have been temporarily paused.

2. What do you think are the major challenges especially with regard to tribal communities in Madhya Pradesh?

Ans: The main challenge is economic instability. Tribal communities rely on traditional practices and struggle to maintain their land and livelihoods. Government policies often hinder their growth, making it difficult for them to achieve financial stability and progress.

3. How is your NGO addressing these challenges?

Ans: We address these issues through income support programs, awareness initiatives, and by preserving tribal culture and religion. We engage directly with the communities, as we share a similar cultural background, which helps build trust. While some external religious movements have tried, they haven't been as successful, particularly in Madhya Pradesh.

4. Where do your financial resources come from?

Ans: Our financial resources primarily come from the Central Government's Tribal Ministry and the Social Welfare Ministry. We also receive contributions from the communities we work with. Additionally, we are seeking CSR funding for community health initiatives. We apply for and work on various projects as they become available, including funding from the Tribal Ministry and HID.

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Questionnaire 4

(Dr. Sumer Singh (ex DC principal, large farm in house of Lalpur, Founder of MU20), (telephonic)

Location: Chandigarh

1. How long have you been practising farming?
   Ans: I have been practising farming for more than 10 years.

2. What are the various crops that you grow?
   Ans: We grow multiple crops on our farms such as Poplar, apple, avocado, peach, plum, moringa, wheat, paddy, bamboo etc.

3. What are the various methods and sources you use for irrigation?
   Ans: For irrigation in our farms we divert water from the Sutlej river through tube wells while we also draw water from groundwater.

4. According to you, what are the most modern and suitable techniques for irrigation in your area?
   Ans: In our area, the most modern techniques for irrigation we use are drip irrigation, sprinklers and as mentioned before we use tube wells for water diversion. Although these techniques are most suitable, they are not very reliable and we still face water scarcity.

5. Are you aware of any government programs or schemes aimed at improving water access in your community?
   Ans: Yes, we are aware of multiple schemes but the smaller farmers in our area aren't aware. let alone the schemes introduced by the central government, even local schemes specially created only for the local farmers of Punjab are also not provided. Not to mention crop specific schemes are also denied.

6. Can you share any personal experiences related to water scarcity?
   Ans: When I first came to Punjab, at that time I saw water scarcity at its peak in villages but thankfully one of my past students was in one of the highest posts in the irrigation department therefore I could get some water arranged at that time but even now every summer there is water scarcity.

7. Have you ever heard of the concept of fog catcher and its working?
   Ans: Yes, I have heard of the concept of fog catchers and its working, but the concept of using bamboo to make fog catchers is quite new to me.

8. What problems do you think the fog catcher could face if implemented in your area?
   Ans: Well these fog catchers are mostly used in hilly countries like Peru but if we use the model proposed by you, it can be of great use in all regions.

9. What specific benefits do you see in diversifying with bamboo?
   Ans: As we grow bamboo we know its multiple benefits for example bamboo yields good prices, is versatile, replenishes soil and doesn't take a lot of effort to grow.

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Social INTERACTIONS



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Visit to Tillore Khurd

AIM: To uncover challenges rural farmers face: water scarcity, lack of technical knowledge, and unawareness of government aid, seeking potential solutions.



- To gather comprehensive insights for our project, we conducted multiple surveys across various stages of the research. Each survey was designed with specific objectives in mind, targeting different respondent groups to ensure broad representation and to capture diverse perspectives.

- To gain a deeper understanding of the challenges faced by farmers at the grassroots level, we visited Tillore village in Khurd. Known for its significant farming community, approximately 757 of the village's 1200 families are involved in agricultural activities.

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The conversation shed light on the challenges faced by farmers in a rural setting. The village of Tillor Khurd is located on the foothills of Ralamandal.

- Water problems in summer impacts the crop yields due to heavy reliance on precipitation
- Lack of technology that utilises less water



Through our interviews, we got to know that farmers in Tillore primarily cultivate soybeans and wheat. We further enquired about the water scarcity and water issues faced by farmers in the region.

We further asked them about the fog pattern of the region, to which they replied that fog is quite frequently noticed from October to December.

While government schemes like Pradhan Mantri Krishi Sinchai Yojana and Jal Shakti Abhiyan exist, farmers struggle to access or benefit from them due to the absence of guidance on application processes.

Overall, the farmers grapple with inefficient resources, limited exposure to technology, and insufficient information on government schemes, thus battling with complex challenges prevailing in the farming community.

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Visit to Chikhli Village

AIM:To explore and comprehend the multifaceted challenges small farmers face in rural Indore, highlighting issues of inadequate utilities, market access, technological awareness, and the absence of political support.

- **The conversation delved into the challenges faced by small farmers in the rural village of Kajligarh, featuring insights from Mr Ramchandra Kushwaha and several farmers of Chikhali village.**

- **One of the major concerns was the government's irregular provision of essential utilities like water and electricity. Despite various schemes, small farmers felt excluded from benefits, which predominantly favoured larger agricultural entities.**



- **The only source of water for the villagers is the small pool or "talab" built by the villagers themselves, which at times also proves to be a necessity.**

- **The absence of government officers' regular visits and limited awareness about modern farming technologies accentuated the farmers' difficulties. They expressed reluctance to adopt new methods due to fear and a lack of education. Additionally, the absence of regular farmer meetings for planning exacerbated their challenges.**

- **In the summer months, water for both drinking and irrigation purposes takes a halt since the only tank present in the village completely dries up and isn't looked after by the panchayat.**

- **We also found that the farmers already grow bamboo in certain parts of the villages showing that the conditions for growing bamboo are favourable,**

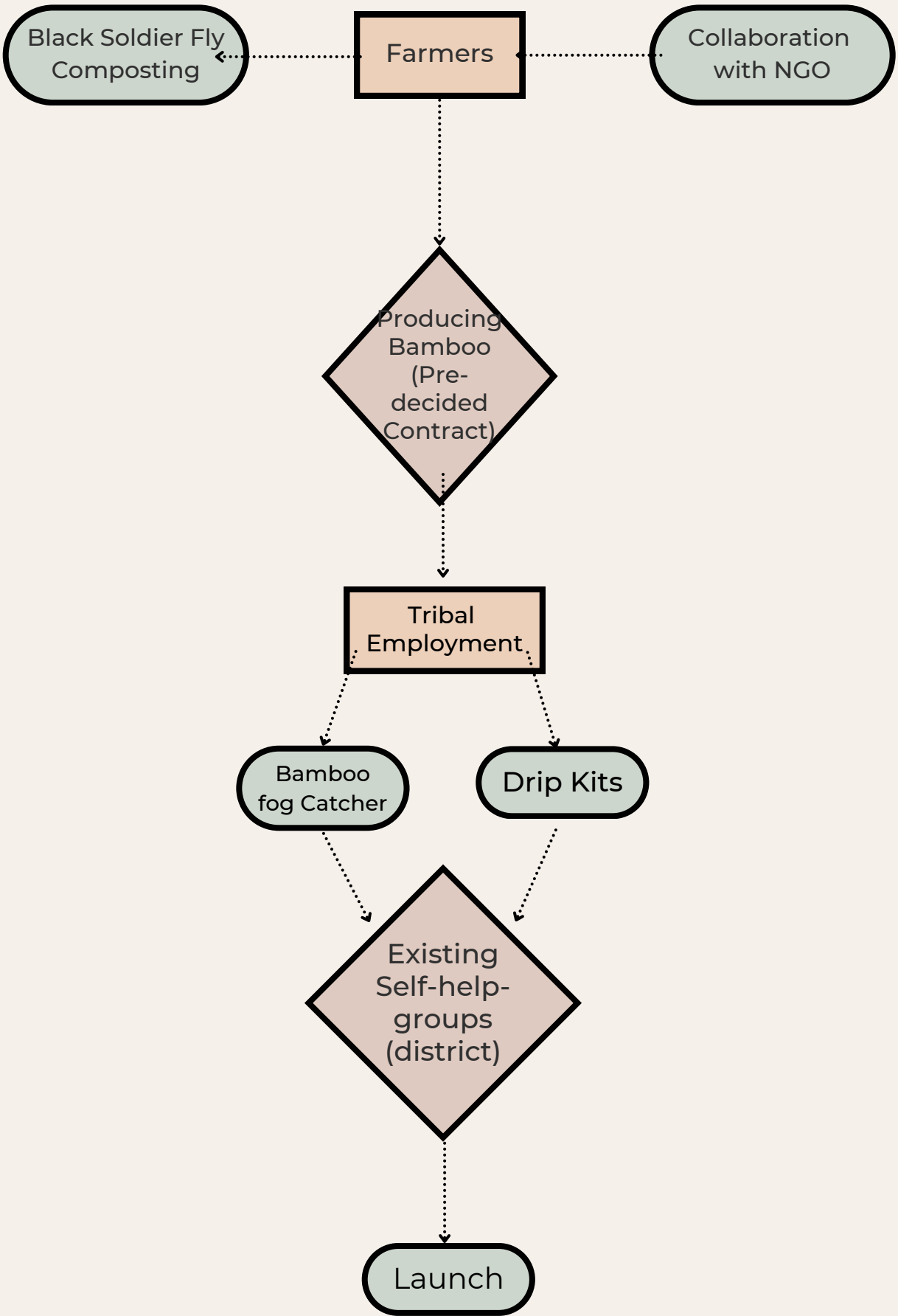
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Visit to Chhappan Dukaan (Nukkad Natak)

- We performed this street play to increase awareness among the crowd at Jhalaria village and Chhappan Dukaan food hub for tribal welfare and their problem of water scarcity.



OUR
SOLUTION STRUCTURE

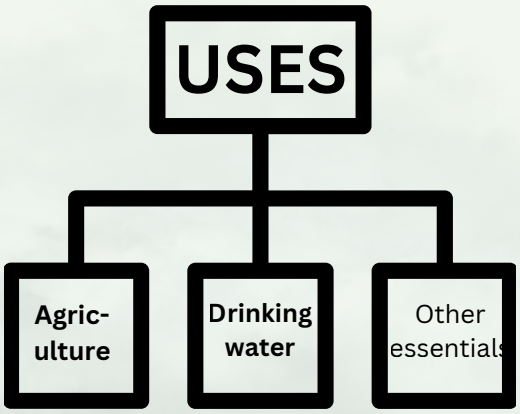


SOLUTIONS EMPHASISED



1. The Bamboo Fog Catcher

A fog catcher made of bamboo fibre is an innovative, eco-friendly solution designed to capture water from fog in arid or water-scarce regions.



Steps for working

1. Fog passes through bamboo fiber nets
2. Tiny droplets of water condense on the fibers
3. Hydrophilic bamboo fibers capture moisture

BENEFITS

- Promotes Sustainable Water Management Practices
- Low-Cost, Low-Maintenance Solution
- Reduces Reliance on Traditional Water Source

Environmental benefits

- Rapid growth of bamboo
- Minimal resource requirements
- Eco-friendly alternative

How much water would be condensed using the bamboo fog catcher?

- The fog catcher model proposed for this study is based on a netted bamboo fibre design, chosen for its durability and eco-friendly qualities.
- The net dimensions are 4 metres in width and 3 metres in height, resulting in a total effective surface area of approximately 56.6 square metres.
- The design includes a dense criss-crossing pattern of bamboo fibres with additional support ropes, maximising the surface area exposed to fog droplets.
- High humidity levels ranging from 70% to 90% (Indian Meteorological Department, 2023).
- Fog density of 10-12 grams per cubic metre during peak winter days,which is conducive to substantial fog water collection.
- Temperatures generally range between 10°C to 18°C, further promoting the condensation of moisture onto fog catcher surfaces These factors together create an environment that supports high fog water collection rates.

Estimation of Daily Water Collection:

- Fog water collectors worldwide have demonstrated yields between 5 and 15 litres per
- square metre per day in fog-rich areas (For Janapav, a conservative estimate of 8 litres
- per square metre per day has been applied based on its climate profile.
- Surface area=56.6m^2
- Average daily yield=8litres/m^2

Each fog catcher, therefore, can be expected to collect approximately **452.8 litres** of water per day under favourable conditions and a lower end estimate would give us about **283 litres** of water per day.



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2. DIY Drip Kits

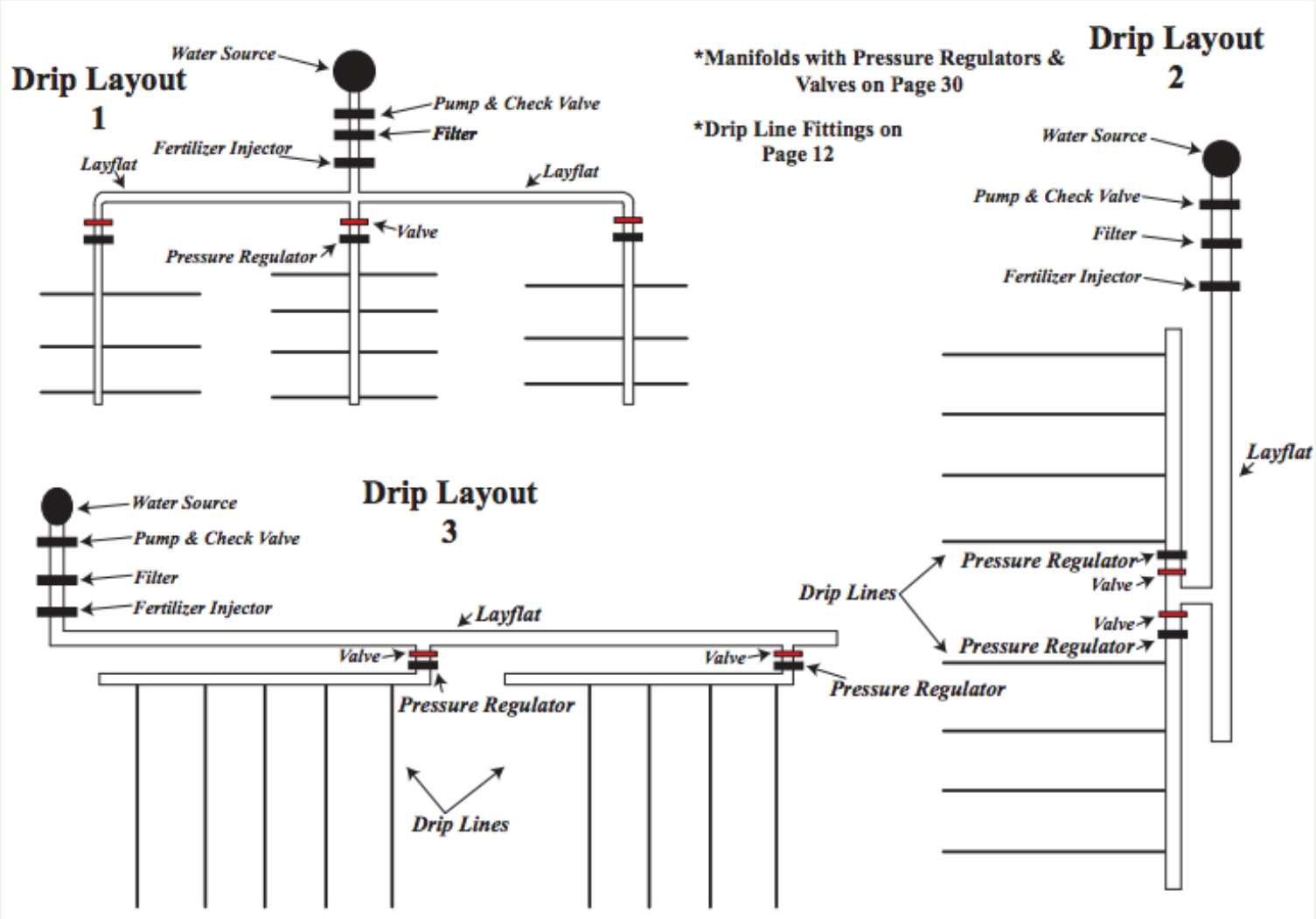


- Bamboo Drip Irrigation Kits is an innovative, sustainable solution designed to address the challenges of water conservation in agriculture, especially in regions with limited access to water.

- By utilising bamboo—an abundant, eco-friendly, and renewable resource—these DIY drip kits allow efficient water delivery directly to plant roots, minimizing water wastage and promoting healthier crop growth.

- First, they reduce water consumption by delivering water directly to the plant's roots through a network of bamboo tubes or pipes, preventing evaporation.
- Additionally, bamboo's natural durability and biodegradability make it a sustainable alternative to plastic-based irrigation systems, helping to reduce environmental pollution.





How will all this be implemented?

1. Source of Raw materials

- Our raw materials are bamboo sticks, poles, fibres, hemp or jute, twine/rope.
- We will acquire raw materials such as bamboo from large-scale farmers. During our survey in the Chikli, Ghosikheda, Pathanpipliya, and Tillorkhurd villages, we interacted with 20 farmers out of which 17 agreed to grow bamboo when offered market price on a predetermined term of the contract. Other materials can be brought in bulk or exported from other states.

2. Production Process

The production process involves tribes in the local areas, like the Baigas, Bansod, Sahari, and Bhil tribes, which work with such materials to make the mesh, nets, baskets, etc. The tribes will be contacted and offered employment through the Samdarshi Seva Kendra NGO, Aastha Foundation, and Sewa Bharti, which work on a large scale with marginalised communities and tribes.

Benefit to tribes:

These NGOs will uplift tribals, especially women, by providing them with employment, healthcare, vocational training, and education, empowering individuals to lead better and more fulfilling lives.

3. Final Implementation

The Drip Kits will be provided to the already existing SHGs in the district. The water will be put in a reservoir and connected to farmers' wells and personal dams through tubes and pipes. The tribes will be employed every year to maintain the fog catchers. The respective panchayats will look over the distribution system.

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2. The Black Soldier Fly Larvae



The Solution

As an additional solution, we also suggest the Black Soldier Fly Larvae (BSFL).Considering in our minds the continuous increase in organic waste, the BSFL is a great and even better solution than existing ones.

What are BSFL?

Black Soldier Flies (BSF) efficiently recycle organic waste, turning agricultural by products into valuable resources. Their larvae provide farmers with an eco-friendly waste management solution and an additional income source, making BSF a sustainable and profitable option for agriculture.

Cost and finance

In India, the price for live BSFL is between Rs 150 and Rs 1000, depending upon the supplier. Through this, we can understand that this method is comparatively affordable for most and gives considerable results.

BSFL SURVIVAL CONDITIONS

1 Humidity

Madhya Pradesh's 40-70% humidity is ideal for BSFL.BSF larvae thrive at 25-35°C, suitable for India, and survive between 15-47°C.

2 Temprature

The BSF larvae generally grow between 25 and 35 degrees Celsius, which is favourable in India, though they can easily survive between temperatures of 15 and 47 degrees.

3 Lighting

The BSFL prefer low light levels generally, therefore instead of getting exposed to sun they can be raised indoors with artificial lighting.

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SO.SCIENCE

Benefits of BSFL

Composting

As the BSFL grow by consuming organic waste, a major part of organic waste can be fed to grow it; this would help in the reduction of organic waste, especially in a country like India.

Feed to animals

As another source of help for the farmers, when these larvae grow very well and in the later stages of their metamorphosis, can be fed to the pet animals as a rich source of proteins such as chickens and others.

Cleaner Environment

Since the organic waste is reduced, we would also be able to see a better environment because multiple greenhouse gases are released into the air due to organic waste.

Water conservation

The BSF is a great method to save water, one of our main aims i.e. they consume 95% less water than traditional methods.

BSF Oil

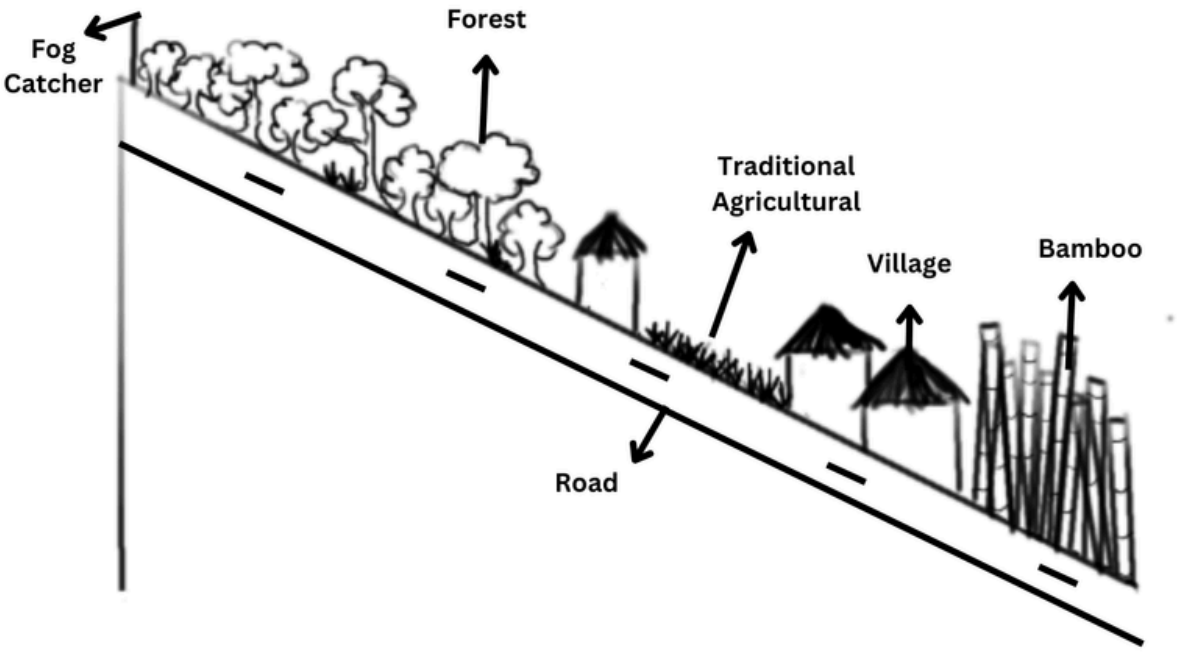
If we get dried BSF larvae, they can be converted into oil that contains high proteins; this oil is sold in the market for high prices.

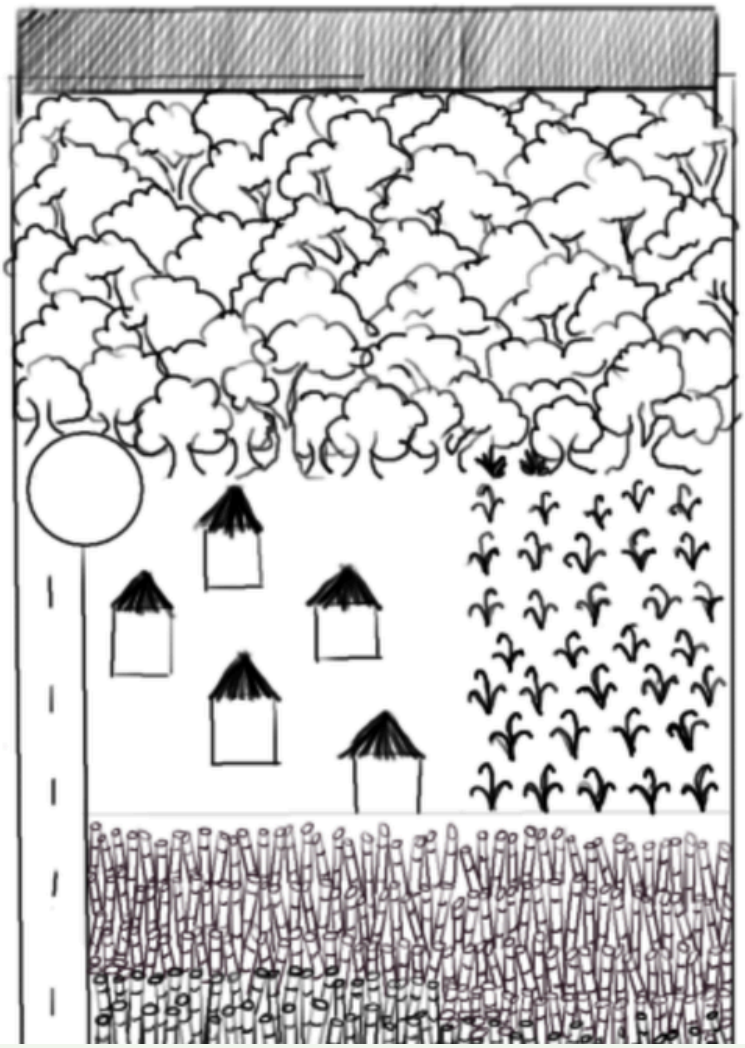
Organic fertilizer

Additionally, when the BSFL consumes organic waste, BSF Frass is formed which is a natural and organic fertiliser. These fertilisers can improve the soil conditions and introduce microorganisms which are beneficial for the soil. This would also reduce the need for artificial fertilisers.

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THE MODEL





Fog CATCHER

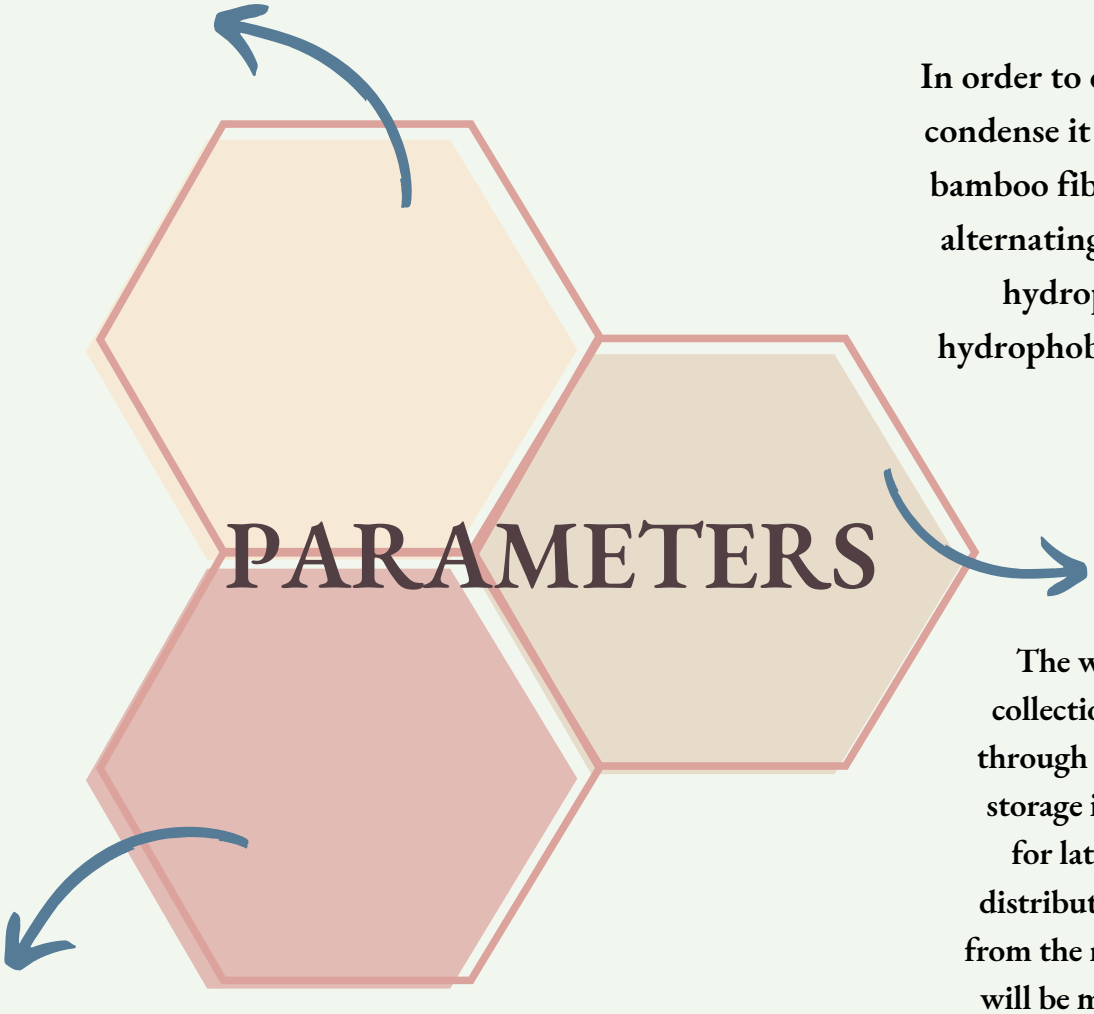
The bamboo fog catchers, designed for hilly areas, capture moisture from fog using bamboo fibres with hydrophilic properties. Then, through the hydrophobic bamboo fibre sheet water gets collected in tanks and is transported down the hill via pipes for future use.

In parallel, the BSF composting system uses the Black Soldier Fly larvae to efficiently process organic waste, converting it into high-quality, nutrient-rich compost. This process not only reduces waste but also, saves water, produces protein-rich larvae that can be used as animal feed and supporting sustainable agriculture.

Black SOLDIER FLY

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In the model of the fog catcher, a wooden plank is kept inclined to depict the hilly regions, where possibilities of fog formation are comparatively more .The measurements of the wooden plank used are 46.5 inches by 30 inches and depict a hill of altitude approximately 300-800 m.



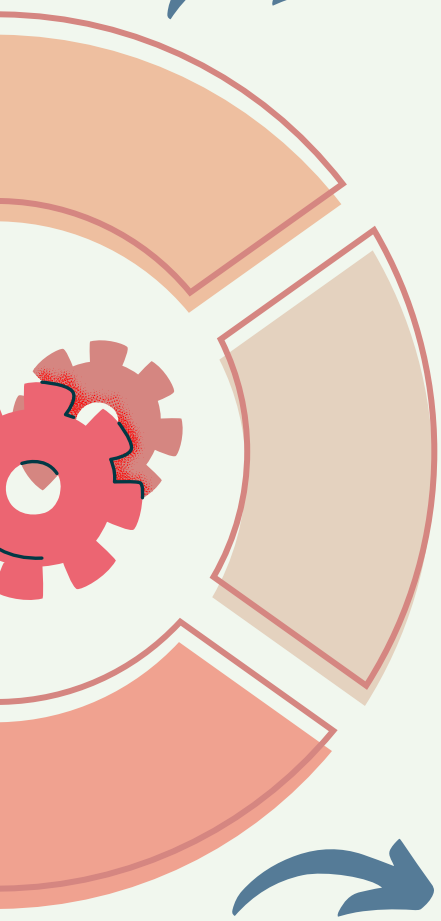
In order to capture fog and condense it into the water, bamboo fibre is used with alternating streams with hydrophilic and hydrophobic properties.

The water after collection will flow through the pipes for storage in reservoirs for later use; the distribution of water from the reservoir tank will be monitored by the gram panchayat.

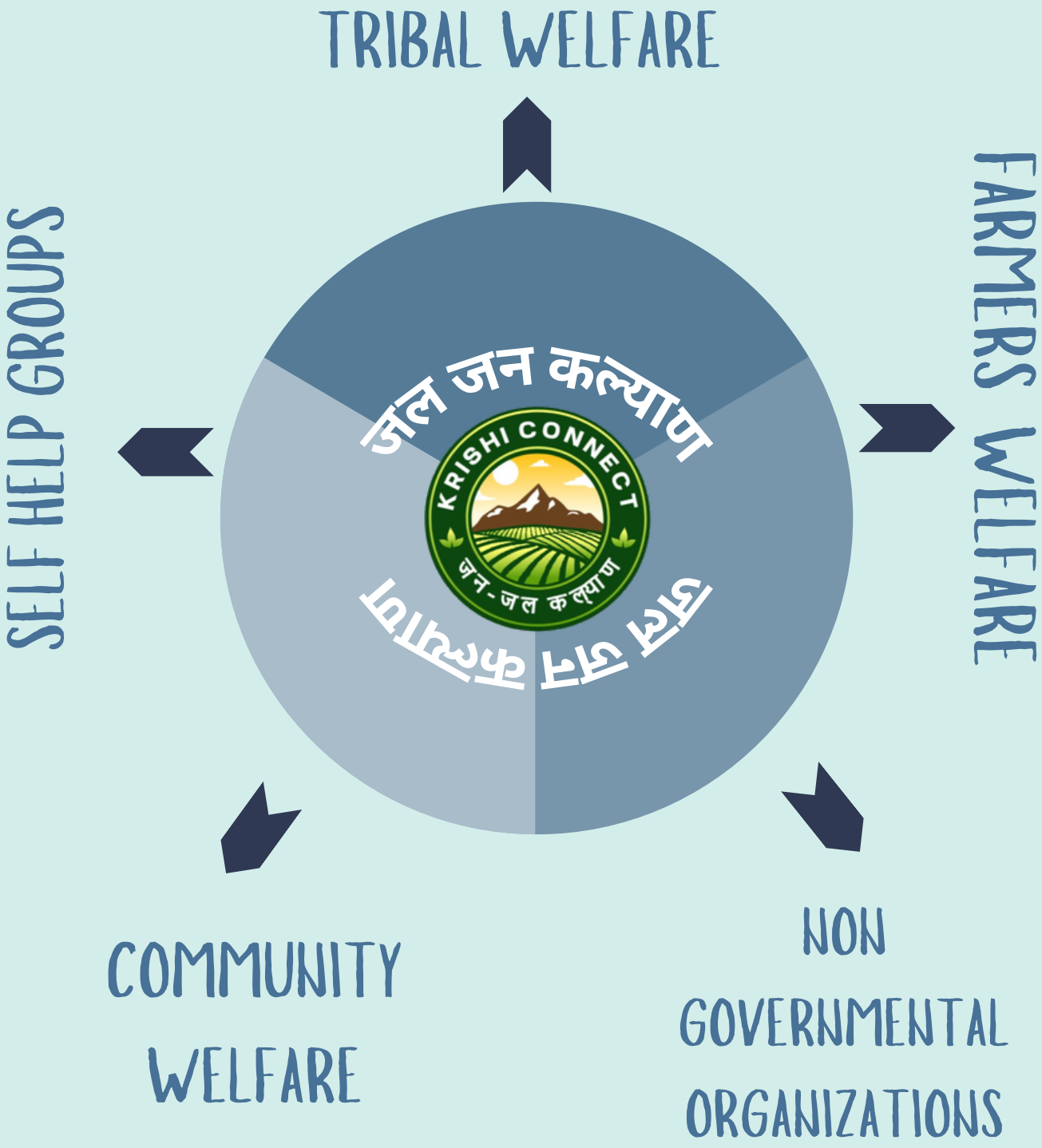
In case of any sort of technical issue regarding the efficiency of the work, the maintenance will be taken care of by the established and trusted self-help groups.

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TECHNOLOGY EMPHASISED



- Use of Hydrophilic Bamboo Fibre
- Sustainable Practices
- Hydrophobic Usage of Bamboo Fibres


TRIBAL WELFARE
FARMERS WELFARE
SELF HELP GROUPS
COMMUNITY WELFARE
NON GOVERNMENTAL ORGANIZATIONS
जल जन कल्याण
KRISHI CONNECT
जन-जल कल्याण

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CONCLUSION

Our project, "Droplets of Hope," aims to help tribal and rural communities thrive by empowering them with sustainable, innovative resource management solutions. We work alongside these communities to improve education, farming, water access, and market connections while respecting their cultural identity.

At KrishiConnect, we're committed to supporting marginalized communities often left behind in development. While facing challenges like limited resources and opportunities, these communities also possess valuable knowledge and traditions.

Inspired by Dr. B.R. Ambedkar’s vision of justice and equality, we believe it's our responsibility to ensure no one is left behind. Through our work, we aim to create a society where everyone can contribute to and benefit from progress.

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